

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	3	4	4	3	2	2	2	2	2	1	3	3	1	3	1	2	3	3	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	3	2	2	3	1	4	1	2	1	2	2	2	3	3	2	3	3	2
Q.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	4	2	2	2	2	2	3	4	3	4	2	4	1	4	4	3	2	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80
A.	4	4	3	1	1	3	2	4	4	1	2	3	2	2	4	2	4	1	1	3
Q.	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	4	1	1	3	3	4	2	2	2	1	1	1	1	3	1	2	2	1	1	2
Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	2	1	3	4	1	1	1	3	1	4	3	3	4	3	1	1	1	1	2
Q.	121	122	123	124	125	126	127	128	129	130										
A.	3	1	4	4	1	3	1	3	1	2										

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Your Hard Work Leads To Strong Foundation

Open

HINT – SHEET

1. Ans (3)

For voltmeter

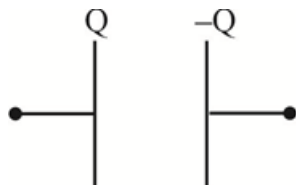
$$R = \frac{V}{I_g} - G$$

$$= \frac{50}{10^{-3}} - 54 \approx 50k\Omega \text{ (A)}$$

For ammeter

$$S = \frac{I_g G}{I - I_g} = \frac{10^{-3} \times 54}{(10 - 1) \times 10^{-3}} = 6\Omega \text{ (C)}$$

2. Ans (3)



$$\text{Charge } Q = CV = \frac{\epsilon_0 A}{2} V = \text{const.}$$

$$W_{\text{ext}} = \Delta U = \frac{Q^2}{2C_f} - \frac{Q^2}{2C_i}$$

$$C_f = \frac{\epsilon_0}{(d/2)}, C_i = \frac{\epsilon_0 A}{d}$$

3. **Ans (4)**

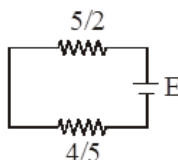
At $t = 0$, current through inductor is zero,

$$\text{hence } R_{eq} = (5+1) \parallel (5+4) = \frac{18}{5}$$

$$i_1 = \frac{E}{18/5} = \frac{5E}{18}$$

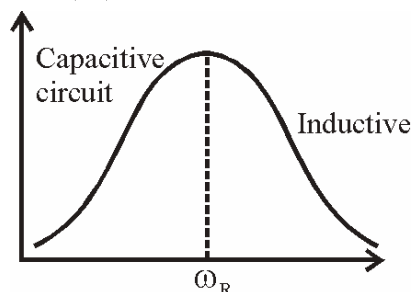
At $t = \infty$, inductor becomes a simple wire and now the circuit will be as shown in figure

$$\text{hence } R_{eq} = (5 \parallel 5) + (4 \parallel 1) = \frac{33}{10} ; (\parallel \Rightarrow \text{parallel})$$



$$i_2 = \frac{E}{33/10} = \frac{10E}{33}$$

4. **Ans (4)**



$$\omega_R = \frac{1}{\sqrt{LC}}$$

$\omega_2 < \omega_R$ from diagram

Circuit will behave as capacitance circuit

$\therefore$ Current leads voltage

5. **Ans (3)**

Use this formula

Diameter = Measured reading – Zero correction

6. **Ans (2)**

Conceptual.

7. **Ans (2)**

Velocity of EM wave is given by $\frac{1}{\sqrt{\mu \epsilon}}$

$$\text{Velocity in air} = \frac{\omega}{k} = c$$

$$\text{Velocity in medium} = \frac{c}{2}$$

$$\therefore \frac{\frac{1}{\sqrt{\epsilon_{r1}}}}{\frac{1}{\sqrt{\epsilon_{r2}}}} = \frac{c}{\left(\frac{c}{2}\right)} = 2$$

$$\Rightarrow \frac{\epsilon_{r1}}{\epsilon_{r2}} = \frac{1}{4}$$

8. **Ans (2)**

Conceptual.

9. **Ans (2)**

For paramagnetic material

According to curies law

$$\chi \propto \frac{1}{T}$$

$$\chi \propto \frac{1}{T} \Rightarrow \chi_1 T_1 = \chi_2 T_2$$

$$\Rightarrow \frac{6}{0.4} \times 4 = \frac{I}{0.3} \times 24$$

$$I = \frac{0.3}{0.4} = 0.75 \text{ A/m}$$

10. **Ans (2)**

$$\text{Average force } F_{av} = \frac{\Delta p}{\Delta t} = \frac{2IA}{c}$$

$$= \frac{2 \times 25 \times 10^4 \times 15 \times 10^{-4}}{3 \times 10^8} = 2.50 \times 10^{-6} \text{ N}$$

11. **Ans (1)**

$$\lambda = \frac{h}{P}, \quad P = \frac{h}{\lambda}$$

12. **Ans (3)**

At $t=0$ disintegration rate = 4250 dpm

At $t=10$ disintegration rate = 2250 dpm

$$A = A_0 e^{-\lambda t}$$

$$2250 = 4250 e^{-\lambda(10)}$$

$$\Rightarrow \lambda(10) = \ln\left(\frac{4250}{2250}\right)$$

$$\Rightarrow \lambda = 0.063 \text{ min}^{-1}$$

13. **Ans (3)**

$$\text{Least count (L.C)} = \frac{0.5}{50}$$

$$\text{True reading} = 5 + \frac{0.5}{50} \times 20 - \frac{0.5}{50} \times 5$$

$$= 5 + \frac{0.5}{50}(15) = 5.15 \text{ mm}$$

Option (3)

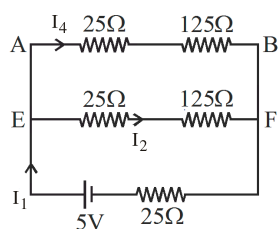
14. Ans (1)

In line CD, diode is in R.B

$$I_3 = 0$$

$$V = i_1 \times R$$

$$5 = i_1 \times 100$$



$$i_1 = \frac{5}{100} \text{ A} = 0.05 \text{ A}$$

$$i_2 = i_4 = \frac{5}{200} \text{ A} = 0.025 \text{ A}$$

15. Ans (3)

$$\text{Velocity of sound } v = \frac{1}{\sqrt{\rho}}$$

$$\frac{v_1}{v_2} = \sqrt{\frac{\rho_2}{\rho_1}} \dots\dots(i)$$

$\therefore$ First overtone of closed organ pipe

$$= \text{First overtone of open organ pipe } \frac{3v_1}{4L_1} = \frac{2v_2}{2L_2}$$

$$L_2 = \frac{4}{3} \frac{v_2}{v_1} L_1 = \frac{4}{3} L \sqrt{\frac{\rho_1}{\rho_2}}$$

($\because L_1 = L$ (given))

16. Ans (1)

Equation of the harmonic progressive wave given $y = a \sin 2\pi(bt - cx)$

$$\text{Here } 2\pi n = w = 2\pi b \Rightarrow v = b$$

$$k = \frac{2\pi}{\lambda} = 2\pi c \Rightarrow \frac{1}{\lambda} = c$$

(Here c is the symbol given for $\frac{1}{\lambda}$ and not the velocity)

$$\therefore \text{Velocity of the wave} = v\lambda = b \left(\frac{1}{c} \right) = \frac{b}{c}$$

$$\frac{dy}{dt} = a2\pi b \cos 2\pi(bt - cx) = a\omega \cos(\omega t - kx)$$

$$\text{Maximum particle velocity} = a\omega = a2\pi b = 2\pi ab \text{ given this is } 2 \times \frac{b}{c}$$

$$\text{i.e., } 2\pi ab = \frac{2b}{c} \text{ or } c = \frac{1}{\pi a}$$

17. Ans (2)

$$\omega = \frac{D\lambda}{d} = \frac{D}{d} \left(\frac{h}{mv} \right)$$

$$\omega \propto \frac{1}{v}$$

18. **Ans (3)**

$$V_{COM} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$= \frac{10 \times 14 + 4 \times 0}{10 + 14} = \frac{14 \times 14}{14} = 10 \text{ m/s}$$

19. **Ans (3)**

$$T = \frac{2\pi}{\omega_1} = \frac{2\pi}{\omega_2} \Rightarrow \omega_1 = \omega_2$$

$$\therefore \frac{\omega_1}{\omega_2} = 1$$

20. **Ans (4)**

Use concept of series combination.

21. **Ans (3)**

$$\text{Rate of flow of water} = A_A V_A = A_B V_B$$

$$(40)V_A = (20)V_B$$

$$V_B = 2V_A \quad \dots\dots (1)$$

Using Bernoulli's theorem

$$P_A + \frac{1}{2}\rho V_A^2 = P_B + \frac{1}{2}\rho V_B^2$$

$$P_A - P_B = \frac{1}{2}\rho(V_B^2 - V_A^2)$$

$$700 = \frac{1}{2} \times 1000(4V_A^2 - V_A^2)$$

$$V_A = 0.68 \text{ m/s} = 68 \text{ cm/s}$$

$$\text{Rate of flow} = A_A V_A$$

$$= (40)(68) = 2720 \text{ cm}^3/\text{s}$$

22. **Ans (3)**

$$g' = g - \omega^2 R$$

23. **Ans (3)**

$$C_p - C_v = \frac{R}{4} = a \quad \dots(1)$$

$$C_p - C_v = \frac{R}{32} = b \quad \dots(2)$$

$$a = 8b$$

24. **Ans (2)**

In the given graph CD represents liquid state.

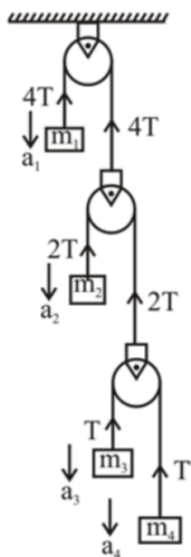
25. **Ans (2)**

When temperature of gas increases, it expands. As the cross-sectional area of right piston is more, therefore greater force will work on it (because $F = PA$). So piston will move towards right.

26. Ans (3)

$$\begin{aligned}\frac{T_1}{T_2} &= \frac{\sin \theta_1}{\sin \theta_2} \\ &= \frac{\sin \theta}{\sin (90 - \theta)} \\ &= \frac{\sin \theta}{\cos \theta} \\ &= \frac{1}{\left(\frac{\cos \theta}{\sin \theta}\right)} = \frac{1}{\cot \theta}\end{aligned}$$

27. Ans (1)



Using constraint

$$\sum \vec{T} \cdot \vec{a} = 0$$

$$-4T a_1 - 2T a_2 - T a_3 - T a_4 = 0$$

$$4a_1 + 2a_2 + a_3 + a_4 = 0$$

28. Ans (4)

$$\begin{aligned}\text{Work} &= \frac{1}{2} I \omega^2 = \frac{1}{2} (1) \frac{(40 \times 10^{-2})^2}{2} \cdot (10 \times 2\pi)^2 \\ &= \frac{1}{2} \cdot 16 \cdot 10^{-2} \cdot 100 \cdot 4\pi^2 = 16\pi^2 = 158\end{aligned}$$

29. Ans (1)

$$I t \propto P^a d^b E^c$$

$$a = -\frac{5}{6}, d = \frac{1}{2}, c = \frac{1}{3}$$

30. Ans (2)

By applying law of conservation of energy

$$mgR = \frac{1}{2} mv^2 \Rightarrow v = \sqrt{2Rg}$$

31. **Ans (1)**

Bakelite- It is thermosetting polymer used for making electrical switches.

Glyptal – manufacture of paints and lacquers

PVC – manufacture of water pipes, rain coats, hand bags

Polystyrene – manufacture of radio and television cabinets

32. **Ans (2)**

2,4-DNP test is useful for the identification of carbonyl compounds.

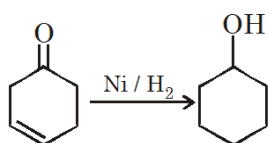
33. **Ans (2)**

Reducing or classical smog is the combination of smoke, fog and SO_2 .

34. **Ans (2)**

$-\text{NO}_2$ and $-\text{Cl}$

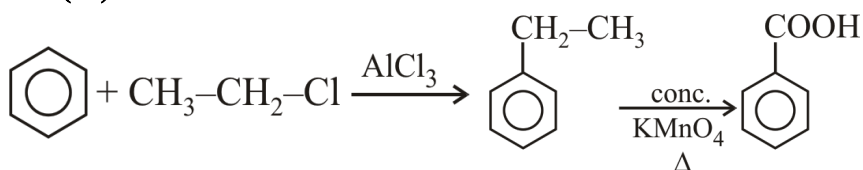
35. **Ans (3)**



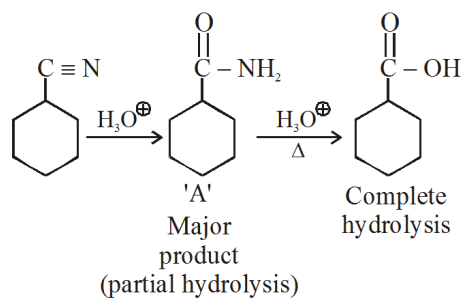
36. **Ans (3)**

Aldol condensation.

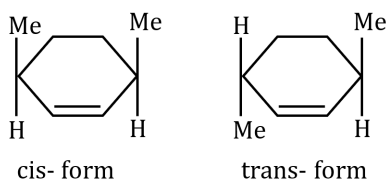
37. **Ans (2)**



38. **Ans (3)**



39. **Ans (3)**



40. **Ans (2)**

Mirror image of D-Galactose is L-Galactose

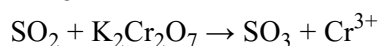
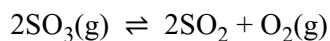
41. **Ans (3)**

Electronic configuration of Rb = $[\text{Ar}] 4s^2 3d^{10} 4p^6 5s^1$

Valence electron in 5s subshell

for 5s, $n = 5$, $l = 0$, $m = 0$, $s = +\frac{1}{2}$ or $-\frac{1}{2}$

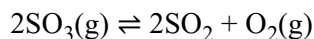
42. **Ans (2)**



$$n_{\text{eq}} \text{ of } \text{SO}_2 = n_{\text{eq}} \text{ of } \text{K}_2\text{Cr}_2\text{O}_7$$

$$2 \times n_{\text{SO}_2} = 6 \times n_{\text{K}_2\text{Cr}_2\text{O}_7}$$

$$n_{\text{SO}_2} = 6 \text{ mole}$$



$$10 \text{ mol}$$

$$10 - x \quad \quad \quad x \quad \quad \quad x/2$$

$$n_{\text{SO}_2} = x = 6$$

$$\text{At eq}^m, \quad 4\text{mol} \quad 6\text{mol} \quad 3\text{mol}$$

$$K_C = \frac{(6/4)^2(3/4)}{(4/4)^2} \Rightarrow \frac{27}{16} \Rightarrow 1.6875 \text{ mol/L}$$

43. **Ans (4)**

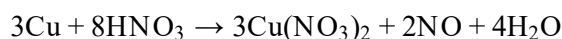
$\therefore$ Reaction is first order

$$\therefore R = k[\text{N}_2\text{O}_5]$$

$$\Rightarrow 2.4 \times 10^{-5} = 3 \times 10^{-5} \times [\text{N}_2\text{O}_5]$$

$$\therefore 0.08 \text{ M}$$

44. **Ans (2)**



$$\begin{aligned} \frac{95.25}{63.5} &= 1.5 \text{ mol} & \frac{8}{3} \times \frac{3}{2} &= 4 \text{ mol} \\ & & &= M \times V (\ell) \\ & & &= 1 \times V (\ell) \end{aligned}$$

$$\therefore V = 4.0 \text{ litre}$$

45. **Ans (2)**

For KCN, Normality = Molarity.

$$K_h = \frac{10^{-14}}{1.6 \times 10^{-9}} = \frac{10^{-5}}{1.6} = \frac{10^{-4}}{16}$$

$$h = \sqrt{\frac{K_h}{C}} = \sqrt{\frac{10^{-4}}{16 \times \frac{1}{100}}} = \sqrt{\frac{10^{-2}}{16}} = \frac{0.1}{4} = 0.025$$

$$= 2.5 \times 10^{-2}$$

46. **Ans (2)**

$$U_{\text{rms}} = \sqrt{\frac{U_1^2 + U_1^2 \dots \dots \dots}{N_1 + N_2 \dots \dots \dots N_a}}$$

$$5 = \sqrt{\frac{4(7)^2 + 6 \times X^2}{10}}$$

$$x = 3$$

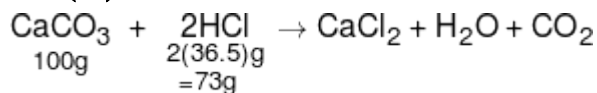
47. **Ans (2)**

$$\pi = CRT$$

$$1.23 = \frac{0.6}{M} \times 10 \times 0.082 \times 300$$

$$M = 119.99$$

48. **Ans (2)**

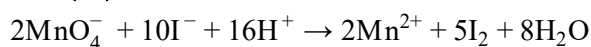


100 g of CaCO_3 react with $\text{HCl} = 73 \text{ g}$

$\therefore$ 5g of CaCO_3 will react with HCl

$$= \frac{73}{100} \times 5 = 3.65 \text{ g}$$

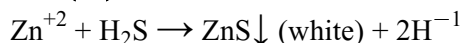
49. **Ans (3)**



50. **Ans (4)**

Size of colloidal particles : 1 nm to 1000 nm.

51. **Ans (3)**



52. **Ans (4)**

(i) In VOSO_4 , 'V' is in +4 oxidation state.

So it act as oxidising agent.

(ii) Cr_2O_3 is an amphoteric oxide.

(iii) In RuO_4 , 'Ru' is in +8 oxidation state.

So it act as oxidising agent.

(iv) Red colour of ruby is due to the presence of Cr^{+3} ions in Al_2O_3 .

53. **Ans (2)**

Calamine : ZnCO_3

Willemite : Zn_2SiO_4

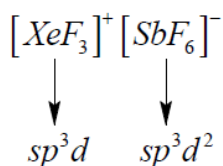
Argentite : Ag_2S

Diaspore : $\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$

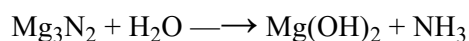
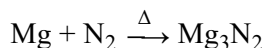
54. **Ans (4)**

Li^+ does not form alum due to its small size.

55. **Ans (1)**



56. **Ans (4)**



57. **Ans (4)**

For d^8 , d^3 , and d^1 electronic configuration in octahedral field, low spin and high spin terms are not defined.

58. **Ans (3)**

K_2CrO_7 (Cr^{+6} does not have any unpaired e^-) is coloured but it is diamagnetic.

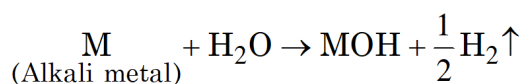
$\Rightarrow \text{Ti}^{+4}$ and Cu^+ do not have any unpaired e^-

$\Rightarrow \text{V}^{+4} \text{OSO}_4$ (V^{+4} has one unpaired e^-) is coloured and paramagnetic.

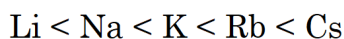
59. **Ans (2)**

oxalate is formed on reaction with sodium formate.

60. **Ans (4)**



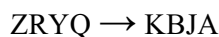
Reactivity order with water



71. **Ans (2)**

Second is the process of gradual disappearances of the first.

72. **Ans (3)**



$$\text{Z} + \text{A} = 26 + 1 = 27$$

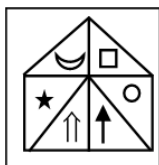
$$\text{R} + \text{J} = 18 + 10 = 28$$

$$\text{Y} + \text{B} = 25 + 2 = 27$$

$$\text{Q} + \text{K} = 17 + 11 = 28$$

$\therefore$ Following same for PWOV we get code for PWOV as FLEK.

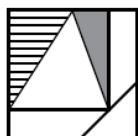
73. **Ans (2)**



In first figure from second, the designs inside the pentagon change their places and the design ' $\uparrow$ ' rotates 90° in clockwise direction and a new design comes at the place on Δ and design ' $\bullet$ ' or ' $\blacksquare$ ' comes without colour.

74. **Ans (2)**

By observation



75. **Ans (4)**

By observation.



76. **Ans (2)**

First words are country in the second except Canberra : Australia.

77. **Ans (4)**

The second is the multiple of the sum of digits of first except 26 : 156.

78. **Ans (1)**

Option (1) because of four pins.

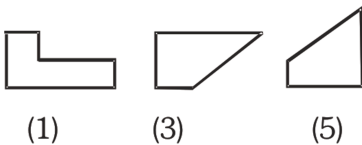
79. **Ans (1)**

By observation



80. **Ans (3)**

By observation



81. **Ans (4)**

Portfolio

82. **Ans (1)**

The sum of all numbers when divided by 3 resulting a number at the centre as,

$$(51 + 11 + 22 + 21)/3 = 35$$

$$\text{and } (10 + 51 + 2 + 24)/3 = 29$$

$$\text{Similarly } (45 + 18 + 63 + x)/3 = 45 \Rightarrow 126 + x = 135 \Rightarrow x = 9$$

Hence, missing number is 9.

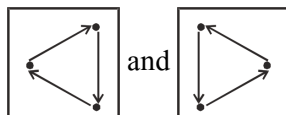
83. **Ans (1)**

$$9^2 + 10, (19)^2 + 20, (29)^2 + 30, (39)^2 + 40, (49)^2 + 50.$$

$$\text{So, ?} = (59)^2 + 60 = 3541$$

84. **Ans (3)**

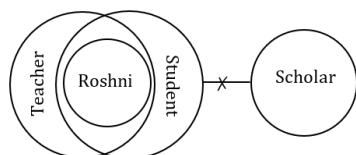
The symbols move in the sequences



and

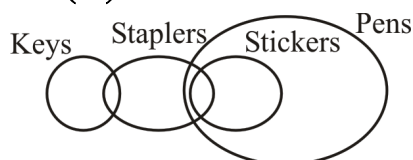
alternately.

85. **Ans (3)**



Clearly, both conclusions follows.

86. **Ans (4)**



Conclusion (1) & (4) are true and either (2) or (3) follows.

87. **Ans (2)**

By observation,



88. **Ans (2)**

By observation



89. **Ans (2)**

It is mentioned in the passage that self-medication helps prevent and treat minor ailments that do not require medical attention.

90. **Ans (1)**

The fact is evident from the sentence 'The Government is likely to further restrict foreign exchange needed for imports' in the passage.

91. **Ans (1)**

$$x = \sec^2 t \Rightarrow \frac{dx}{dt} = 2 \sec t \cdot \sec t \cdot \tan t$$

$$y = \cot t \Rightarrow \frac{dy}{dt} = -\operatorname{cosec}^2 t$$

$$\frac{dy}{dx} = -\frac{1}{2}(\cot t)^3 \Rightarrow \left(\frac{dy}{dx} \right)_{t=\frac{\pi}{4}} = -\frac{1}{2}$$

$$\text{Equation of tangent } y - 1 = -\frac{1}{2}(x - 2)$$

$$x + 2y = 4 \quad \dots(A)$$

$$\text{Equation of curve } x - \frac{1}{y^2} = 1$$

$$\Rightarrow y^2 = -\frac{1}{1-x} \quad \dots(B)$$

from (A) and (B)

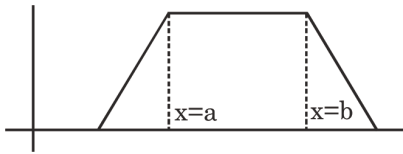
$$2y^3 - 3y^2 + 1 = 0$$

$$\text{for point Q } y = -\frac{1}{2}, x = 5$$

$$\text{Distance } |PQ| = \sqrt{(5-2)^2 + \left(-\frac{1}{2} - 1\right)^2} = \frac{3\sqrt{5}}{2}$$

92. **Ans (1)**

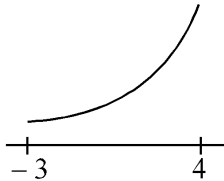
Ans. (1)



Graph of $f(x) \Rightarrow$ infinite points of maxima

93. **Ans (1)**

(A) $f(x)$ has no relative minimum on $(-3, 4)$



(B) $f(x)$ is continuous function on $[-3, 4]$

$\Rightarrow f(x)$ has min. and max. on $[-3, 4]$ by IVT

(C) $f''(x) > 0$

$\Rightarrow f(x)$ is concave upwards on $[-3, 4]$

(D) $f(3) = f(4)$

By Rolle's theorem

$\exists c \in (3, 4)$, where $f'(c) = 0$

$\Rightarrow \exists$ critical point on $[-3, 4]$

94. **Ans (3)**

$$\lim_{h \rightarrow 0} f(1+h) \leq f(1)$$

$$-2 + \log_2(k^2 - 4) \leq 1 - 1 + 10 - 7$$

$$\log_2(k^2 - 4) \leq 5$$

$$k^2 - 4 \leq 2^5 \text{ and } k^2 - 4 > 0$$

$$k \in [-6, -2) \cup (2, 6]$$

95. **Ans (1)**

$$4\sin 2\theta + a(1 + \cos 2\theta) = a + 5$$

$$\Rightarrow 4\sin 2\theta + a\cos 2\theta = 5$$

$$-\sqrt{a^2 + 16} \leq 5 \leq \sqrt{a^2 + 16}$$

$$\Rightarrow a^2 + 16 \geq 25$$

$$\Rightarrow a^2 \geq 9$$

$$\therefore a \in (-\infty, -3] \cup [3, \infty)$$

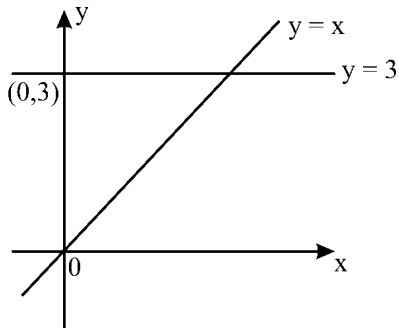
96. **Ans (2)**

Let $f(x) = y$

$$\therefore y^2 - 4y + 3 \geq 0 \Rightarrow (y-1)(y-3) \geq 0 \quad \dots(1)$$

$$y^2 - 6y + 8 \leq 0 \Rightarrow (y-4)(y-2) \leq 0 \quad \dots(2)$$

$$y^3 - 5y^2 + 10y - 12 \leq 0 \Rightarrow (y-3)(y^2 - 2y + 4) \leq 0 \quad \dots(3)$$



from (1),(2) & (3), we get $f(x) = 3$ i.e. constant function

$$A = \frac{1}{2} \times 3 \times 3 \Rightarrow A = \frac{9}{2}$$

97. **Ans (2)**

take $x^{\log_b a} = a^{\log_b x} = y$

$$(a^{\log_b x})^2 - 5 \cdot x^{\log_b a} + 6 = 0$$

$$\Rightarrow y^2 - 5y + 6 = 0$$

$$\Rightarrow (y-2)(y-3) = 0$$

$$\Rightarrow y = 2, 3$$

$$x^{\log_b a} = 3$$

$$\Rightarrow \log_b a \cdot \log_3 x = \log_3 3$$

$$\Rightarrow \log_3 x = \frac{1}{\log_b a} = \log_a b$$

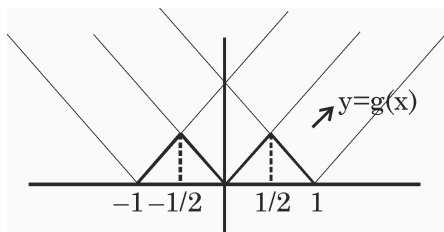
$$x = 3^{\log_a b} \text{ and similarly } x = 2^{\log_a b}$$

98. **Ans (1)**

$$\int_0^3 (1+x)^{20} dx = \int_0^3 ({}^{20}C_0 + {}^{20}C_1 x + \dots + {}^{20}C_{20} x^{20}) dx$$

99. **Ans (1)**

Points of Non-differentiability is 3



From the graph

100. Ans (2)

(I) $x = -1$ (common negative root)

$$\Rightarrow p - q + r = 0$$

(II) $x = 2, 3$

(III) $a = \pm 2$

(IV) $p = 1$

101. Ans (2)

$$|z| = |z - 4| \Rightarrow x = 2 \quad \dots(1)$$

$$(2 + i)z + (i - 2)\bar{z} - 4i = 0$$

$$\Rightarrow x + 2y = 2 \quad \dots(2)$$

by (1) & (2)

$$(x = 2, y = 0)$$

Let centre of circle 'C' i.e. (2, 0) and radius 'R' is equal to $(x - 2)^2 + y^2 = R^2$

So by condition of orthogonal with $x^2 + y^2 = 1$

$$\therefore R = \sqrt{3}$$

102. Ans (2)

(P) common chord is $S_1 - S_2 = 0$

$$\Rightarrow 2x + 2y - 5 = 0$$

$$\Rightarrow 2x + 2y = 5$$

$$\therefore a = 2, b = 1$$

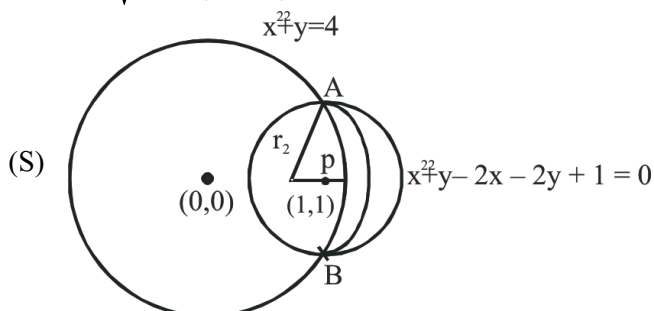
(Q) $\cos \theta = \frac{r_1^2 + r_2^2 - d^2}{2r_1r_2}$,

d is distance between centres

$$\cos \theta = \frac{4 + 1 - 2}{2 \times 2 \times 1} \Rightarrow \cos \theta = \frac{3}{4}$$

(R) These circles have two direct common tangents

$$L = \sqrt{d^2 - (r_1 - r_2)^2} = \sqrt{2 - 1} = 1$$



Equation of AB is $2x + 2y - 5 = 0$

AB is diameter of smallest circle

$$\Rightarrow \text{diameter} = \text{length of AB}$$

$$= 2\sqrt{r_2^2 - p^2} = 2\sqrt{1 - \frac{1}{8}} = \sqrt{\frac{7}{2}}$$

103. Ans (1)

$$\begin{aligned}
 (1 + \ln x) \frac{dx}{dy} - x \ln x &= e^y \\
 \left. \begin{aligned} x \ln x &= t \\ (1 + \ln x) \frac{dx}{dy} &= \frac{dt}{dy} \end{aligned} \right\} \Rightarrow \frac{dt}{dy} - t &= e^y \\
 \Rightarrow te^{-y} &= \int e^{-y} \cdot e^y dy + c \\
 t &= ye^y + e^y \cdot c \\
 x \ln x &= ye^y + e^y \cdot c \\
 0 &= c \\
 \Rightarrow x \ln x &= ye^y \\
 \ln x^x &= y e^y \\
 x^x &= e^{ye^y}
 \end{aligned}$$

104. Ans (3)

$$\begin{aligned}
 \frac{d}{dx} \left(\frac{f(x)}{x^2} \right) &= \frac{x^2 f'(x) - f(x) 2x}{x^4} \\
 \text{by wrong calculation } \frac{d}{dx} \left(\frac{f(x)}{x^2} \right) &= \frac{f'(x)}{2x} \\
 \frac{f'(x)}{2x} &= \frac{xf'(x) - 2f(x)}{x^3} \\
 \Rightarrow x^2 f'(x) &= 2x f'(x) - 4 f(x) \\
 \Rightarrow \frac{f'(x)}{f(x)} &= \frac{4}{2x - x^2} = 2 \left[\frac{1}{x} + \frac{1}{2 - x} \right]
 \end{aligned}$$

On integrating

$$\ell n f(x) = 2[\ell nx - \ell n(2 - x)] + \ell n C$$

$$f(x) = C \cdot \frac{x^2}{(2 - x)^2} ; \lim_{x \rightarrow \infty} \frac{Cx^2}{(2 - x)^2} \Rightarrow C = 4$$

$$\therefore f(x) = \frac{4x^2}{(2 - x)^2}$$

$$\begin{aligned}
 \text{Also } \sum_{r=1}^{n+2} \frac{r^2}{f(r)} &= \frac{r^2(r-2)^2}{4r^2} = \frac{1}{4} \sum_{r=1}^{n+2} (r-2)^2 \\
 &= \frac{1}{4} \left(1 + \frac{n(n+1)(2n+1)}{6} \right)
 \end{aligned}$$

105. Ans (4)

$$\begin{aligned}
 &\int_0^{1.5} x[x^2] dx \\
 &\int_0^1 0 dx + \int_1^{\sqrt{2}} x dx + \int_{\sqrt{2}}^{1.5} 2x dx \\
 &\left[\frac{x^2}{2} \right]_1^{\sqrt{2}} + [x^2]_{\sqrt{2}}^{1.5} \\
 &\left(\frac{2}{2} - \frac{1}{2} \right) + (2.25 - 2) \\
 &\frac{1}{2} + \frac{1}{4} = \frac{3}{4}
 \end{aligned}$$

106. Ans (1)

$$D = \begin{vmatrix} 2 & -4 & \lambda \\ 1 & -6 & 1 \\ \lambda & -10 & 4 \end{vmatrix}$$

$$= 2(3\lambda + 2)(\lambda - 3)$$

$$D_1 = -2(\lambda - 3)$$

$$D_2 = -2(\lambda + 1)(\lambda - 3)$$

$$D_3 = -2(\lambda - 3)$$

When $\lambda = 3$, then

$$D = D_1 = D_2 = D_3 = 0$$

$\Rightarrow$ Infinite many solution

when $\lambda = -\frac{2}{3}$ then D_1, D_2, D_3 none of them is zero so equations are inconsistent $\therefore \lambda = -\frac{2}{3}$

107. Ans (1)

If $P(x, y)$ divides the line $A(x_1, y_1)$ & $B(x_2, y_2)$

in $m : n$ ratio and eccentricity $e = \sqrt{1 - \frac{b^2}{a^2}}$

where is

for the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

we know length = ℓ

$$\text{so } x^2 + y^2 = \ell^2$$

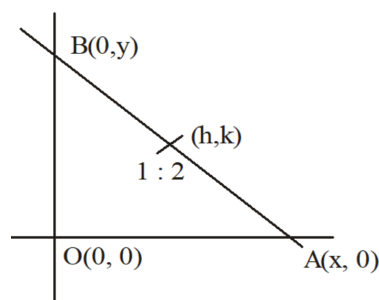
$p(h, k)$ is

$$\left(\frac{x}{3}, \frac{2y}{3} \right)$$

$$\text{so } h = \frac{x}{3}, k = \frac{2y}{3}$$

$$x = 3h \quad y = \frac{3}{2}k, \quad \text{an ellipse } 9h^2 + \frac{9k^2}{4} = \ell^2$$

$$e = \frac{\sqrt{3}}{2}$$



108. Ans (1)

$$\lim_{x \rightarrow e} \frac{1}{[\ln x]}$$

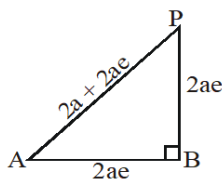
Here we have to take only the right hand limit because when $x < e^-$ is not in the domain

$$\Rightarrow \lim_{x \rightarrow e^+} \frac{1}{[\ln x]}$$

$$x = e^t$$

$$\lim_{t \rightarrow 1} \frac{1}{[t]} = 1$$

109. Ans (3)



When $AP = 2a + 2ae$

$$\Rightarrow AP^2 = AB^2 + PB^2$$

$$\Rightarrow 4a^2 + 4a^2e^2 + 8a^2e = 4a^2e^2 + 4a^2e^2$$

$$\Rightarrow e^2 - 2e - 1 = 0$$

$$\Rightarrow e = \frac{2 + 2\sqrt{2}}{2} \text{ or } \frac{2 - 2\sqrt{2}}{2} \text{ Reject.}$$

$$\Rightarrow e = 1 + \sqrt{2} \Rightarrow k = 1 + \sqrt{2}$$

$$\alpha = 67\frac{1}{2}^\circ \text{ and } 247\frac{1}{2}^\circ$$

$$\text{Sum of all values} = \frac{7\pi}{4}$$

$$a = 7 \text{ \& } b = 2$$

110. Ans (1)

$$\int \frac{x^2(x + x^2 + x^3)}{\sqrt{15x^4 + 12x^5 + 10x^6}} dx$$

$$= \frac{x^2}{30} \sqrt{15x^4 + 12x^5 + 10x^6} + c$$

$$\therefore f(x) = x^2 \therefore f''(1) + \lambda = 32$$

111. Ans (4)

$$\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2},$$

$$\text{when } x \in [-1, 1]$$

112. Ans (3)

Put $\sin x = t$

$$\text{when } x \rightarrow \frac{\pi}{2} \quad t \rightarrow 1$$

$$\text{Now solve } \lim_{t \rightarrow 1} \frac{t - t^t}{1 - t + \ln t}$$

113. **Ans (3)**

$$n^p = p\lambda + n \quad \text{for } p \in \text{prime}$$

$$\lambda \in \text{prime}$$

$$n \in \text{prime}$$

$$\text{so, } n^{11} = 11\lambda_1 + n;$$

$$n^7 = 7\lambda_2 + n;$$

$$n^5 = 5\lambda_3 + n; n^3 = 3\lambda_4 + n; \lambda_1, \lambda_2, \lambda_3, \lambda_4 \in \mathbb{Z}$$

$$\begin{aligned} \text{Now, } E &= \frac{11\lambda_1 + n}{11} + \frac{7\lambda_2 + n}{7} + \frac{5\lambda_3 + n}{5} + \frac{2(3\lambda_4 + n)}{3} - \frac{116n}{1155} \\ &= (\lambda_1 + \lambda_2 + \lambda_3 + 2\lambda_4) + n \left[\frac{1}{11} + \frac{1}{7} + \frac{1}{5} + \frac{2}{3} - \frac{116}{1155} \right] \\ &= \lambda_1 + \lambda_2 + \lambda_3 + 2\lambda_4 + n \left(\frac{1271}{1155} - \frac{116}{1155} \right) = \text{Integer} \end{aligned}$$

114. **Ans (4)**

$$(p \vee \sim r) \vee (\sim p \wedge q \wedge r) = (p \vee \sim r) \vee (\sim (p \vee \sim r) \wedge q) = (p \vee \sim r) \vee q = \sim (\sim p \wedge \sim q \wedge r)$$

115. **Ans (3)**

$$A = \begin{bmatrix} x^2 + 2x & 2x + 1 & 1 \\ 2x + 1 & x + 2 & 1 \\ 3 & 3 & 1 \end{bmatrix} \quad \text{and } \det(A) = (x - 1)^3$$

$$B = \begin{bmatrix} 0 & (2x + 1) & 1 \\ -(2x + 1) & 0 & 3 \\ -1 & -3 & 0 \end{bmatrix} \quad \text{and } \det(B) = 0$$

$$\therefore \det(\text{adj}A) = (x - 1)^6$$

$$\text{or } (x - 1)^6 = 729$$

$$\text{or } x = 4$$

116. **Ans (1)**

$$n = \left(\frac{1}{7} \times \frac{7}{9} \right) + \left(\frac{3}{7} \times \frac{8}{9} \right) + \left(\frac{2}{7} \times \frac{5}{9} \right) + \left(\frac{1}{7} \times \frac{8}{9} \right)$$

$$m = \frac{1}{7} \times \frac{7}{9}$$

$$\text{Required probability} = \frac{1}{7}$$

117. **Ans (1)**

$$\begin{aligned} \text{Maximum number of triangles} &= {}^{18}C_3 - 3 {}^6C_3 \\ &= 756 \end{aligned}$$

118. Ans (1)

equation of tangent at (6, 9) is $x \cdot 6 = 2(y + 9)$

$$\Rightarrow 3x - y - 9 = 0$$

Equation of circle is :

$$(x - 6)^2 + (y - 9)^2 + \lambda(3x - y - 9) = 0$$

$\therefore$ eqn of circle is :

$$(x - 6)^2 + (y - 9)^2 + 10(3x - y - 9) = 0$$

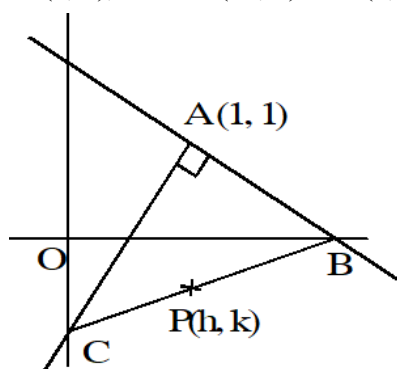
$$62 + 82 + \lambda(0 - 1 - 9) \Rightarrow \lambda = +10$$

$\therefore$ equation of circle is ;

$$\Rightarrow x^2 + y^2 + 18x - 28y + 27 = 0$$

119. Ans (1)

Let $P \equiv (h, k)$, So $B \equiv (2h, 0)$ $C \equiv (0, 2k)$



Clearly as $AC \perp AB$; $\frac{2k-1}{0-1} \times \frac{0-1}{2h-1} = -1$

$$\Rightarrow 2k - 1 = 1 - 2h. \text{ So, locus is } x + y = 1$$

120. Ans (2)

$$x^3 + 2x^2 - 4x + 5 = (x - \alpha)(x - \beta)(x - \gamma)$$

Put $x = 2$ we get $13 = (2 - \alpha)(2 - \beta)(2 - \gamma)$

$$\text{Now } \alpha^3 + 5 = -2\alpha^2 + 4\alpha = 2\alpha(2 - \alpha)$$

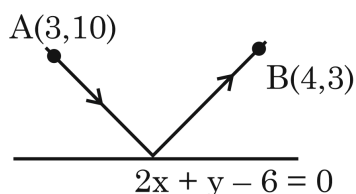
$$\text{So } \frac{(\alpha^3 + 5)(\beta^3 + 5)(\gamma^3 + 5)}{\alpha\beta\gamma}$$

$$= \frac{2\alpha(2 - \alpha)2\beta(2 - \beta)2\gamma(2 - \gamma)}{\alpha\beta\gamma}$$

$$= 8(2 - \alpha)(2 - \beta)(2 - \gamma)$$

$$= 8 \times 13 = 104$$

121. Ans (3)



$$4 + 3 \cdot 3 - \lambda = 0 \Rightarrow \lambda = 13$$

122. Ans (1)

$$n(F \cup H \cup B) = 500 - 50 = 450$$

$$= n(F) + n(H) + n(B) - n(F \cap H)$$

$$- n(F \cap B) - n(H \cap B) + n(F \cap H \cap B)$$

$$\Rightarrow 450 = 285 + 195 + 115 - 45 - 70 - 50$$

$$+ n(F \cap H \cap B)$$

$$\Rightarrow n(F \cap H \cap B) = 20$$

$$n(\text{Exactly one}) = n(F) + n(H) + n(B) - 2 \times n(F \cap H) - 2 \times n(H \cap B) - 2 \times n(F \cap B) + 3 \times n(F \cap H \cap B)$$

$$= 285 + 195 + 115 - 2(45 + 70 + 50) + 3 \times 20$$

$$= 325$$

123. Ans (4)

$$a_{50} - a_{49} = \frac{3}{5} = d$$

$$\& T_{15} = a_n - 14d = 100 - 14 \times \frac{3}{5} = \frac{458}{5}$$

124. Ans (4)

$$10 = \frac{7 + 10 + 11 + 15 + a + b}{6}$$

$$\Rightarrow a + b = 17 \quad \dots(i)$$

$$\frac{20}{3} = \frac{7^2 + 10^2 + 11^2 + 15^2 + a^2 + b^2}{6} - 10^2$$

$$a^2 + b^2 = 145 \quad \dots(ii)$$

Solve (i) and (ii) $a = 9, b = 8$ or $a = 8, b = 9$

$$|a - b| = 1$$

125. Ans (1)

Any plane through the line $x + y + z + 3 = 0$ $3x + y - z$ will be

$$3x + y - z + \lambda (x + y + z + 3) = 0$$

$$(3 + \lambda)x + (1 + \lambda)y + (\lambda - 1)z + 3\lambda = 0 \quad \dots(1)$$

this plane is parallel to the line

$$\frac{x-2}{1} = \frac{y+3}{2} = \frac{z-5}{3}$$

$$\Rightarrow (3 + \lambda)1 + (1 + \lambda)2 + (\lambda - 1)3 = 0$$

$$6\lambda + 2 = 0; \quad \lambda = -\frac{1}{3}$$

Put value of λ in

$$8x + 2y - 4z - 1 = 0$$

$\Rightarrow$ Direction ratios of normal to this plane

$$8 : 2 : -4 = 4 : 1 : -2$$

126. Ans (3)

$$\frac{1}{\sin\left(\frac{\pi}{4} + x\right)} + \frac{1}{\sin\left(\frac{\pi}{4} - x\right)} = 2\sqrt{2}$$

$$\frac{\sqrt{2}}{\cos x + \sin x} + \frac{\sqrt{2}}{(\cos x - \sin x)} = 2\sqrt{2}$$

$$\frac{\cos x}{\cos^2 x - \sin^2 x} = 1$$

$$\cos x = 2\cos^2 x - 1$$

$$2\cos^2 x - \cos x - 1 = 0$$

$$2\cos^2 x - 2\cos x + \cos x - 1 = 0$$

$$\cos x = 1, \cos x = -\frac{1}{2}$$

$$\lambda_1 = 0, \lambda_2 = \frac{2\pi}{3}$$

$$\sin^2 \lambda_1 + \sin^2 \lambda_2 = 0 + \frac{3}{4}$$

127. Ans (1)

Given $t_1 + t_2 + t_3 = 5$

$$t_1 t_2 t_3 = 1$$

and $x + y + z = \frac{3\pi}{4}$

$$\frac{\sum t_i - t_1 t_2 t_3}{1 - (\sum t_i t_j)} = -1$$

$$\Rightarrow 5 - 1 = \sum t_i t_j - 1$$

$$\Rightarrow \sum t_i t_j = 5$$

$$t^3 - 5t^2 + 5t - 1 = 0 \quad \{t_1; t_2; t_3\}$$

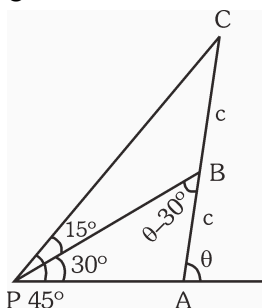
$$t = 1, 2 + \sqrt{3}, 2 - \sqrt{3}$$

$$x = \frac{\pi}{4}, y = \frac{5\pi}{12}, z = \frac{\pi}{12}$$

$$y - z = \frac{\pi}{3}$$

128. Ans (3)

Applying m - n theorem of trigonometry, we get



$$(c + c) \cot(\theta - 30^\circ) = c \cot 15^\circ - c \cot 30^\circ$$

$$\text{or } \cot(\theta - 30^\circ) = 1$$

$$\Rightarrow \theta - 30^\circ = 45^\circ$$

$$\therefore \theta = 75^\circ$$

129. Ans (1)

$$\vec{a} + \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b}) + 2(\vec{b} \cdot \vec{c}) + 2(\vec{c} \cdot \vec{a}) = 0$$

$$\lambda = \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \frac{-3}{2}$$

$$\vec{d} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$$

$$\Rightarrow \vec{a} \times \vec{b} = \vec{b} \times \vec{c} = \vec{c} \times \vec{a}$$

$$\Rightarrow \vec{d} = 3(\vec{a} \times \vec{b})$$

130. Ans (2)

$$\vec{OP} \perp \vec{OQ}$$

$$\Rightarrow -x + 2y - 3x = 0$$

$$\Rightarrow y = 2x \quad \dots(i)$$

$$|\vec{PQ}|^2 = 20$$

$$\Rightarrow (x+1)^2 + (y-2)^2 + (1+3x)^2 = 20$$

$$\Rightarrow x = 1$$

$\vec{OP}, \vec{OQ}, \vec{OR}$ are coplanar

$$\Rightarrow \begin{vmatrix} x & y & -1 \\ -1 & 2 & 3x \\ 3 & z & -7 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 2 & -1 \\ -1 & 2 & 3 \\ 3 & z & -7 \end{vmatrix} = 0$$

$$\Rightarrow 1(-14 - 3z) - 2(7 - 9) - 1(-z - 6) = 0$$

$$\Rightarrow z = -2$$

$$\therefore x^2 + y^2 + z^2 = 1 + 4 + 4 = 9 \quad \text{Option (2)}$$